

Digitally Controlled Analog Music Mixing Console

Skylar Denno

Evan Forcucci

Xinhao Chen

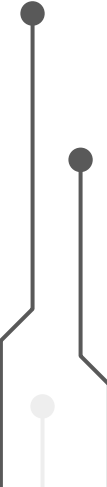
Daniil Petrov

Daniel Salas

Luke Garrity

Professor Siami, Team #2

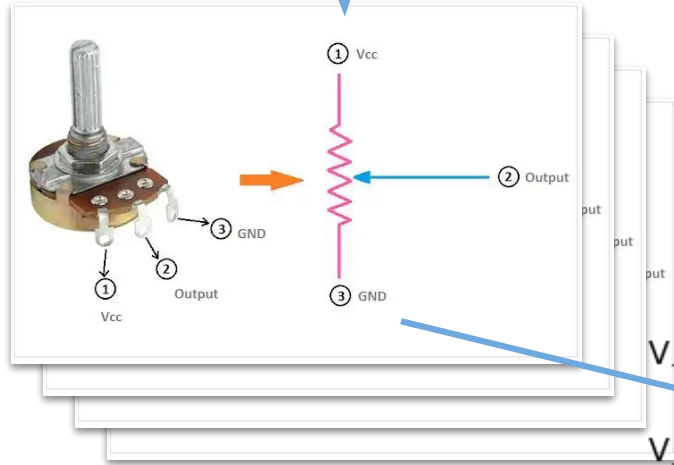
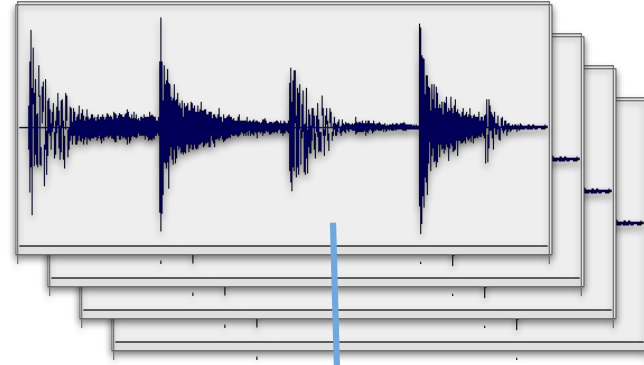




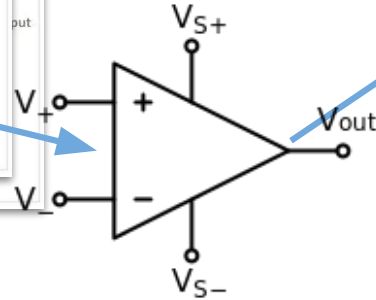
References:
Mixer [1]
s Window [2]

Traditional

x16-64



x100-1000



References:
Waveform [3]
Potentiometer [4]
Op Amp [5]
NS-10s [6]

Project: _____ **Artist:** _____
Date: _____ **Engineer:** _____ **#** ____ **of** _____
Estudio: _____ **Assistant:** _____

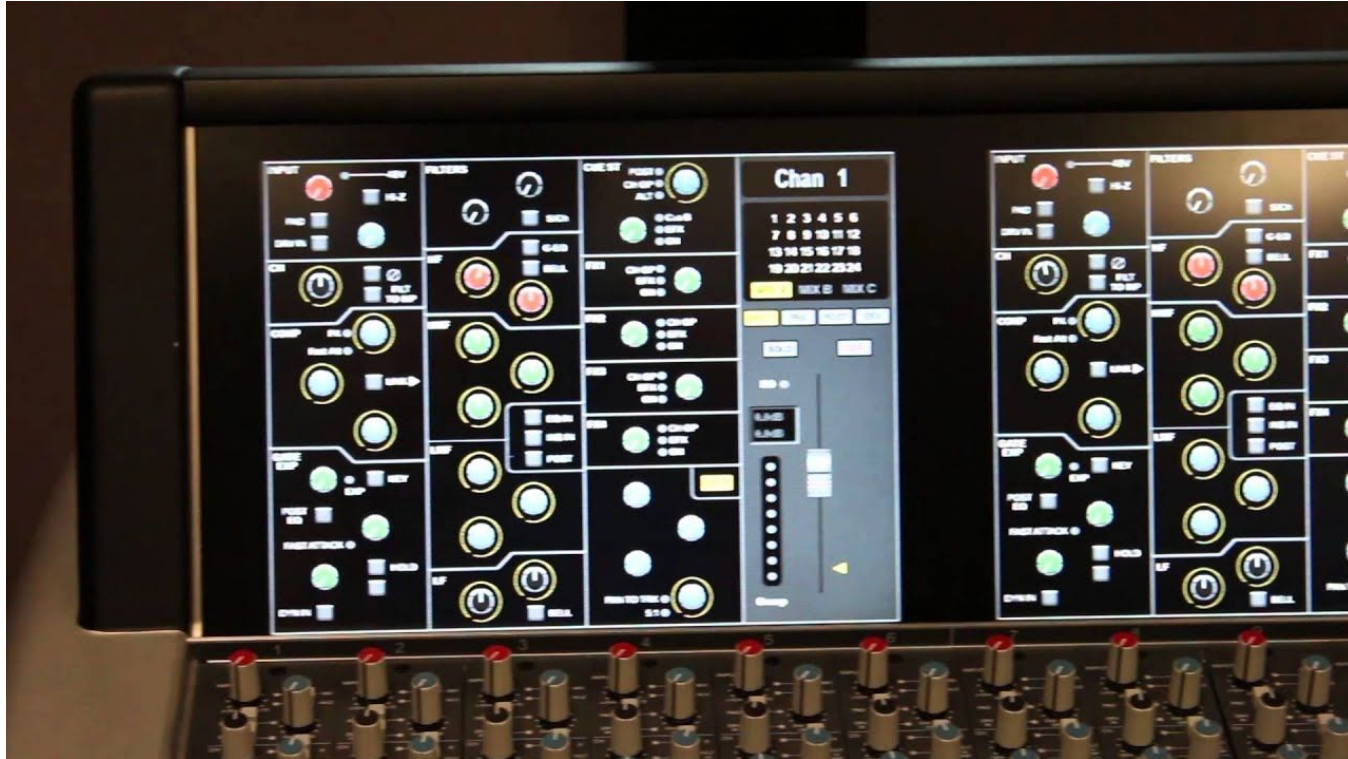
CH		CH		CH		CH	
MIC		MIC		MIC		MIC	
LINE		LINE		LINE		LINE	
HPF		HPF		HPF		HPF	
LF		LF		LF		LF	
HPF 2		HPF 2		HPF 2		HPF 2	
LMP		LMP		LMP		LMP	
LF 2		LF 2		LF 2		LF 2	
DIRECT		DIRECT		DIRECT		DIRECT	
POST		POST		POST		POST	
CHS ST		CHS ST		CHS ST		CHS ST	
FX1		FX1		FX1		FX1	
FX2		FX2		FX2		FX2	
REC		REC		REC		REC	
MIX		MIX		MIX		MIX	
FDR		FDR		FDR		FDR	

CH	FRM	CH LINE IN	INSERT	FRM GP1	TAPE MON IN Divergence
49					
50					
51					
52					
53	GP25	DOMAIN 1 Return $\frac{1}{2}$			
54	GP26				
55	GP27	DOMAIN 2 Return $\frac{1}{2}$			
56	GP28				
57					
58		Front L			
59		Symphony I/O Return R			
60		Rear L			
61		Rear R			
62					
63					
64	ST. 9	Altiverb - Live Chambers			
SEND	AUX PATCHES	RETURN	GRP	PATCH	RETURN
ST. CUE	PTIN 5-6 ➡ Altiverb - Live Chambers	2		SG	
AUX 1	PTIN 2 ➡	3		SG	
AUX 2	PTIN 4 ➡ Altiverb -	4		SG	
AUX 3	PTIN 3 ➡ Altiverb - Apogee Amb.	5		H3000	FX Rtn Front
AUX 4	LFE	6		H3500 #1	FX Rtn Rear
CUE 5		7			
		8			
STEREO MIX BUS PATCHES					
Comp Side-Chain DN410					
Mix Bus PreVCA					
Mix Bus PostVCA					
ADDITIONAL CUES MASTER					
				PT DLY CHLN 53 CHLN 54 CHLN 55 Apogee FX Rear	
				PCM 70 #1 PCM 70 #2 Left Front Right Front Center Sub Group Left Rear Right Rear Center	
CUE-AUX SENDS 		STEREO ECHO RETURNS 		QUAD COMPRESSOR 	
CUE ST 1 2 3 4		1 2 3 4		1 2 3 4	
NOTES: Pro Tools HD as kHz Multitrack Multitrack I/O = -18 dBfs = 0Vu					

References: Left [7]; Right [8]

Solid State Logic

O X F O R D • E N G L A N D

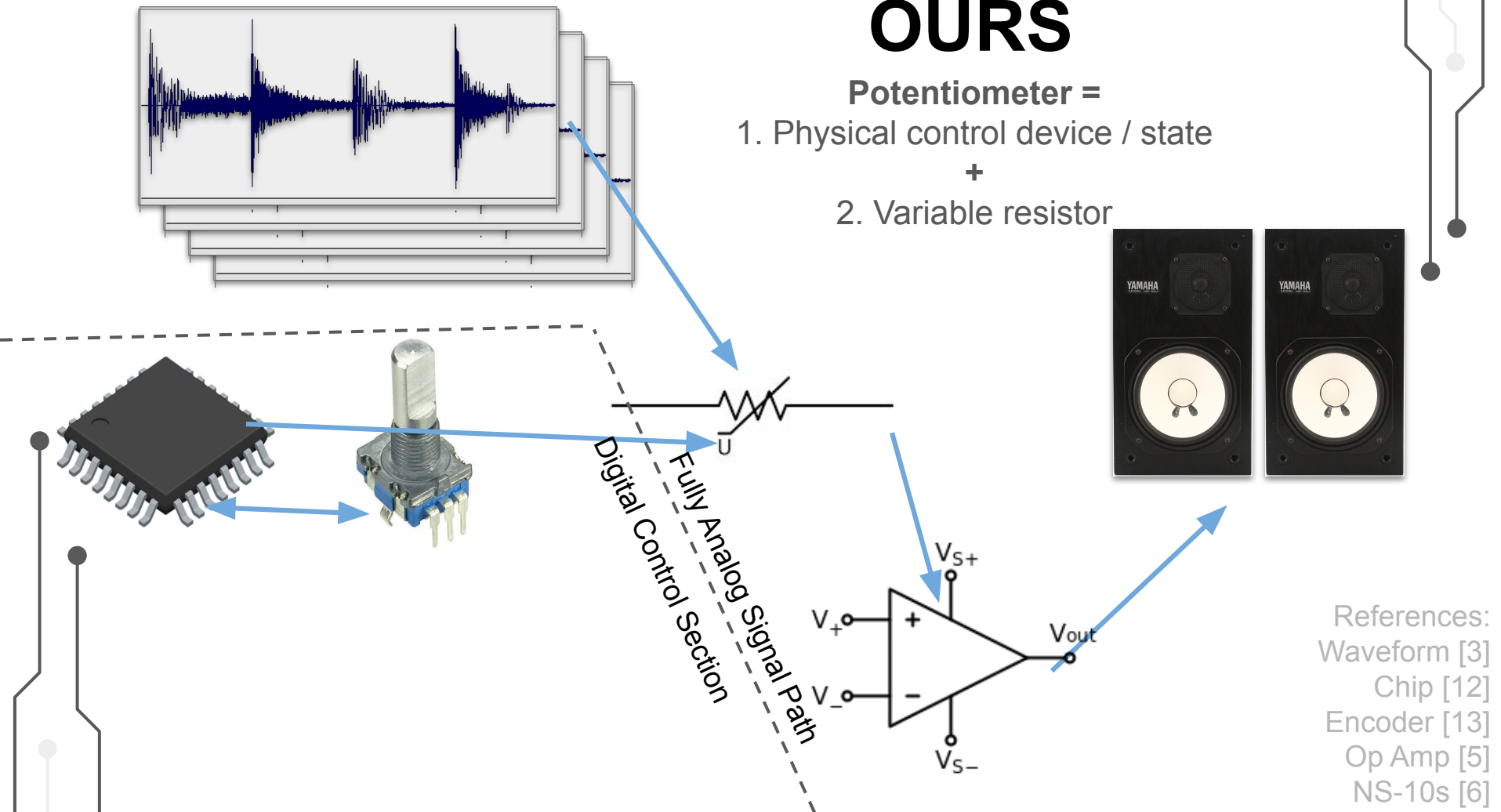


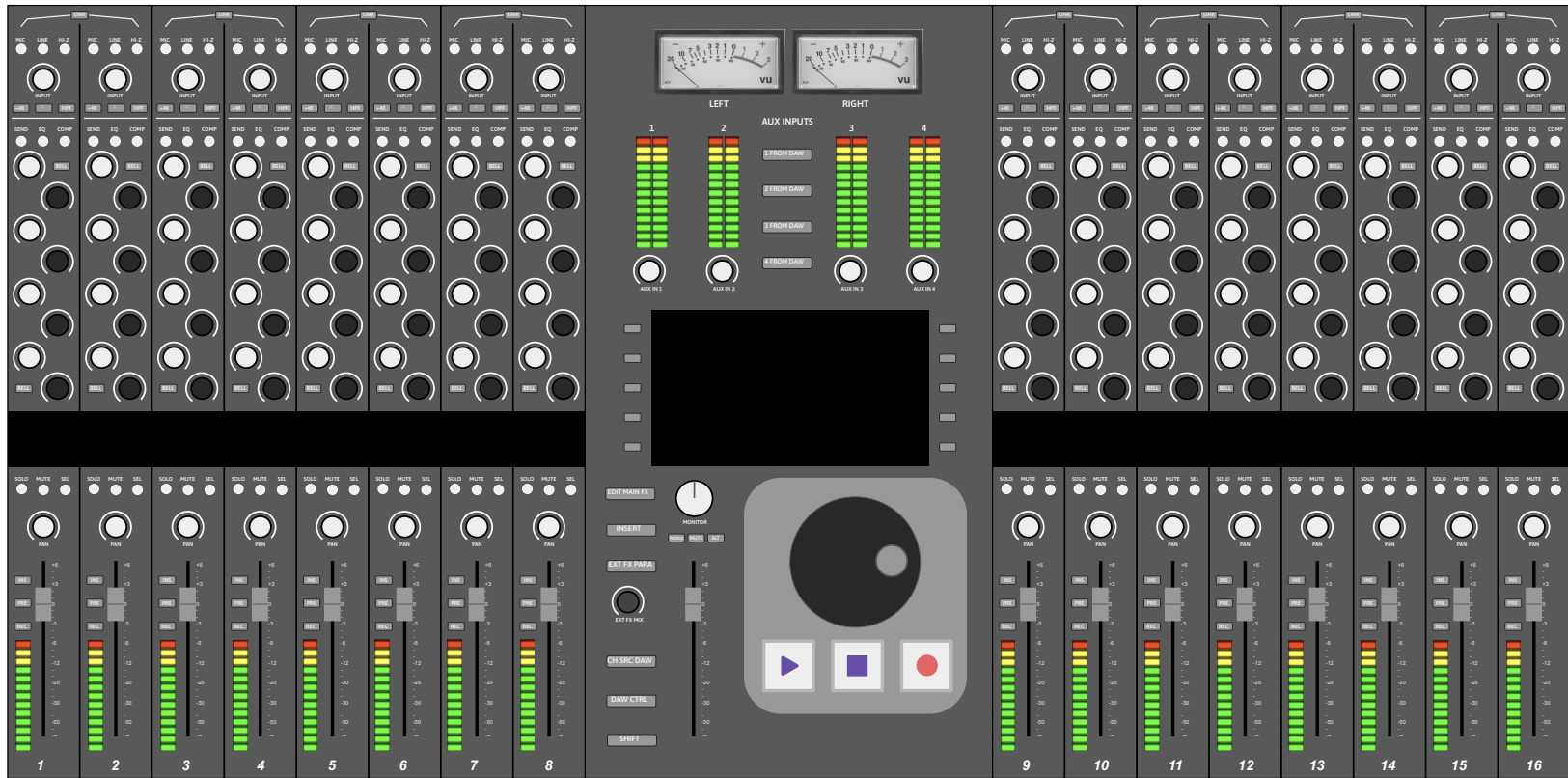
References: SSL Logo [9]; SSL Duality [10]; SSL Duality Recall [11]

OURS

Potentiometer =

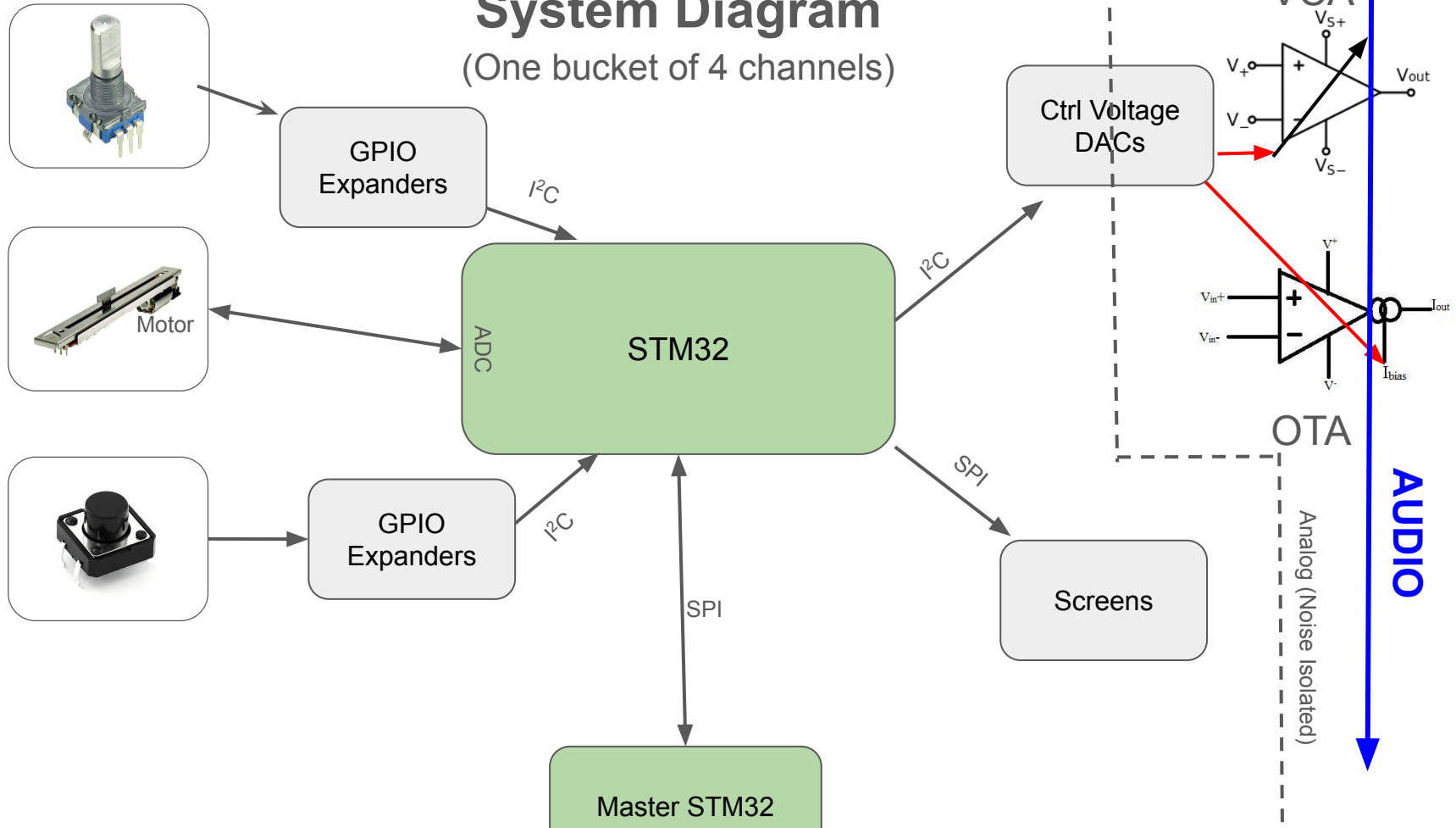
1. Physical control device / state
- +
2. Variable resistor





System Diagram

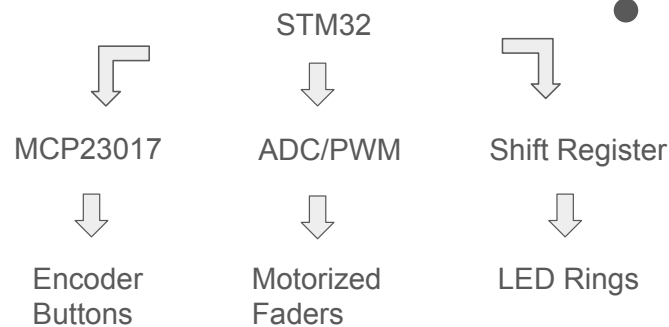
(One bucket of 4 channels)



Control Hardware Integration

Goal: Make console easy and intuitive to use while keeping the actual audio path analog

- Start with software control, then integrate physical controls
- GPIO expansion using MCP23017 over I²C to handle a large number of the buttons and rotary encoders
- Rotary encoders with LED rings for position feedback
- Buttons, motorized faders for tactile control
- When a saved console state is recalled, the faders can automatically move back to the correct position.



Overall: This part focuses on giving the user physical, tactile control while still allowing digital recall and automation

Subteam:
Xinhao & Evan

Digital Embedded Design

- STM32: Probably G474 (Good A/D integration)
 - 1x as main controller + 1x per 4-channel bucket
- ADCs: Just to calibrate and read fader position, not for the audio!
 - 1x per channel
- DACs: Produce the control voltages
 - 20x per channel
- I²C & SPI busses to communicate between chips / devices
- LCD screens for things like track names, EQ transfer curve visualization
 - 1x per channel, 1x large main display

Subteam:
Evan, Xinhao,
Daniil

Firmware

Challenge: Coordinate physical controls, analog control states, and user-interface feedback across the console with low latency while supporting state recall

Approach: A STM32 controller manages a four-channel bucket of physical inputs and analog control voltage states. Another STM32 in the center console collects and distributes recall states. FreeRTOS is used to divide firmware roles into independent tasks

Goals:

- Responsive interrupt- and refresh-based interface IO
- Full-console synchronization and instant setting recall

Subteam:
Daniil & Skylar

Analog Design

Biggest challenge: Preserve audio quality; remote control system is as transparent as possible

- OTAs for voltage controlled resistors
- VCAs for voltage controlled gain

Goals:

- Equalizer w/ state variable filters
- Transparent routing, gain stages
- Repeatable recall; not temp. sensitive

Subteam:
Luke & Skylar

Manufacturing - PCB/CAD

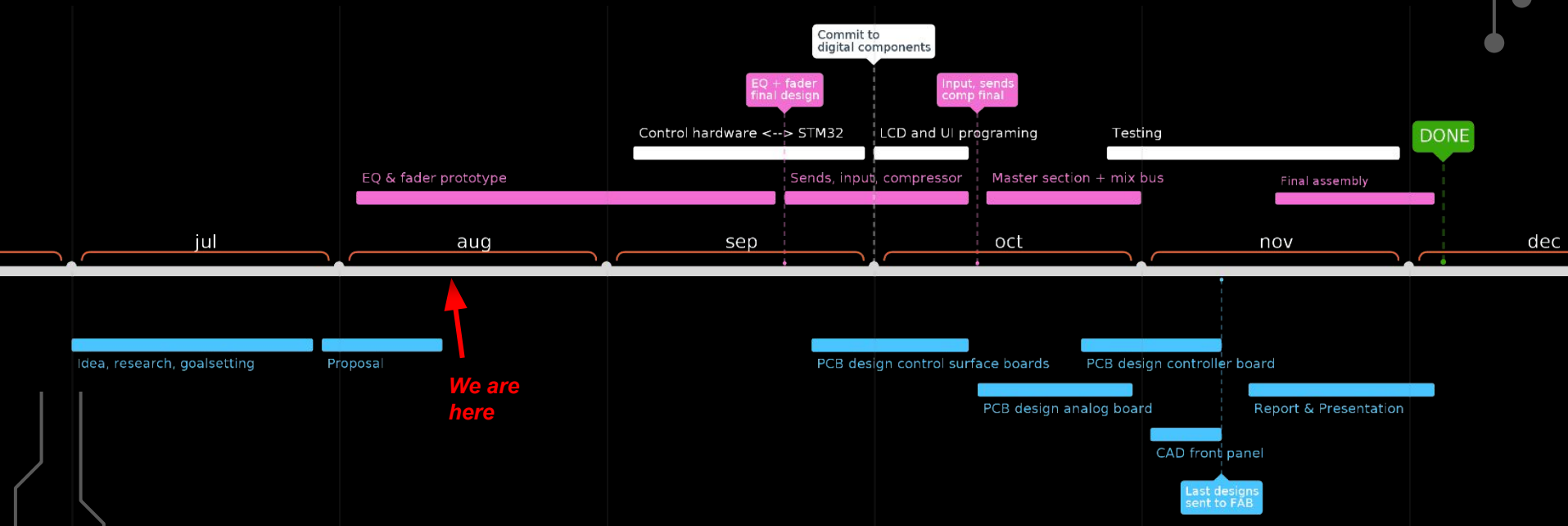
Biggest challenge: NOISE! A lot of controlling it is in the PCB layout

Goals:

- Low noise PCBs for analog stuff: ground planes, power filtering
- High speed (170MHz) PCBs for digital stuff: impedance control, decoupling, crosstalk control
- Create a front panel that makes using the device intuitive

Subteam:
Daniel & Skylar

Timeline



Budget - Bill of Materials

Part	Function	Qty	Unit	Total
<i>Digital control</i>				
STM32G474	Bucket/master MCU	5	\$11.00	\$55.00
MCP23017	GPIO expander	48	\$1.71	\$82.08
MCP4728	Control voltage DAC	80	\$2.07	\$165.60
74HC139	Display CS decoder	4	\$0.21	\$0.84
<i>User interface</i>				
05A20L22P	FFC connector	16	\$0.07	\$1.12
TTP223-BA6	Fader touch sensor	17	\$0.17	\$2.89
DRV8871	Fader H bridge	17	\$2.15	\$36.55
74HCT595	LED shift register	32	\$0.14	\$4.48
AFY800480B1	Master display	1	\$40.86	\$40.86

Held by the team, not budgeted: 17 motorized faders,
160 rotary encoders, 256 illuminated buttons,
16 channel displays, LED ring boards, knob and fader caps.

Unit prices are quantity-break prices at the quantities shown. Excludes shipping, tariff, and sales tax.

Part	Function	Qty	Unit	Total
<i>Analog signal path</i>				
NE5532	Audio op-amp	128	\$0.18	\$23.04
LM13700	Transconductance	64	\$0.75	\$48.00
OPA2134	Summing amp	2	\$1.74	\$3.48
SSI2164	Quad VCA	112	\$0.39	\$43.68
TL074	CV amplifier	64	\$0.21	\$13.44
INA2137	Line receiver	16	\$2.69	\$43.04
DG409	Routing mux	42	\$1.06	\$44.52
THAT1646	Line driver	2	\$6.02	\$12.04
<i>Microphone input</i>				
PGA2500	Mic preamp	16	\$12.83	\$205.28
<i>Fabrication</i>				
—	Front panel	1	\$100.00	\$100.00
—	PCB fabrication	1	\$150.00	\$150.00
—	Passives, diodes	1	\$30.00	\$30.00
Total			\$1,105.94	

Budget - Full Build

Subsystem	Cost
Digital control	\$303.52
User interface	\$85.90
Analog signal path	\$231.24
Microphone input stage	\$205.28
PCB, panel, and passives	\$280.00
Total	\$1,105.94

The microphone input stage is 16 PGA2500 preamplifiers — the single largest line, at 19% of the build.

Excludes shipping, tariff, and sales tax.

Budget - Scaling Down

Subsystem	16 ch	16 ch no mic in	8 ch no mic in
Digital control	\$303.52	\$303.52	\$151.50
User interface	\$85.90	\$85.90	\$67.76
Analog signal path	\$231.24	\$231.24	\$131.66
Microphone input stage	\$205.28	—	—
PCB, panel, and passives	\$280.00	\$280.00	\$280.00
Total	\$1,105.94	\$900.66	\$630.92
With 10% contingency	\$1,216.53	\$990.73	\$694.01

Cost does not scale linearly due to bulk component pricing. PCB and panel costs are uncertain. We will continue to explore alternatives to decrease component costs.

Excludes shipping, tariff, and sales tax.

References

- [1] "KitchenAid 3 Speed Hand Mixer, Empire Red, KHM312," Walmart. Accessed: Aug. 13, 2026. [Online image]. Available: <https://www.walmart.com/ip/KitchenAid-3-Speed-Hand-Mixer-Empire-Red-KHM312/367339369>
- [2] Danny Danyamal Lee, " Creating A Timeline Selection Marker in ProTools," YouTube, Feb. 20, 2017. Accessed: Aug. 13, 2026. [Online image]. Available: <https://www.youtube.com/watch?v=GcEqr9A02ac>
- [3] "PIC MCU: reliably capture and measure drum transients with ADC," Electronics Stack Exchange. Accessed: Aug. 13, 2026. [Online]. Available: <https://electronics.stackexchange.com/questions/506704/pic-mcu-reliably-capture-and-measure-drum-transients-with-adc>
- [4] "Different types of potentiometer," Onzuu. Accessed: Aug. 13, 2026. [Online]. Available: <https://www.onzuu.com/blog/different-types-of-potentiometer>
- [5] "Operational amplifiers," Electronics-Course.com. Accessed: Aug. 13, 2026. [Online]. Available: <https://electronics-course.com/opamp>
- [6] "A history of Yamaha studio reference monitors," Yamaha Music Hub. Accessed: Aug. 13, 2026. [Online]. Available: <https://hub.yamaha.com/proaudio/recording/a-history-of-yamaha-studio-reference-monitors/>
- [7] "SSL AWS 900 channel strip recall sheet," Course Hero. Accessed: Aug. 13, 2026. [Online]. Available: <https://www.coursehero.com/file/119674766/SSL-AWS-900-Channel-Strip-Recall-Sheet-3pdf/>
- [8] "Recall sheets," Wolff Audio. Accessed: Aug. 13, 2026. [Online]. Available: <https://www.wolffaudio.com/support/recall-sheets>
- [9] Solid State Logic, "Solid State Logic company logo," Solid State Logic. Accessed: Aug. 13, 2026. [Online image]. Available: <https://solidstatelogic.com/>
- [10] "SSL Duality Delta 24 channel studio mixing console with DAW control," Studiocare. Accessed: Aug. 13, 2026. [Online image]. Available: <https://studiocare.com/products/ssl-duality-delta-24-channel-studio-mixing-console-with-daw-control-poa>
- [11] Joe Gill, " SSL Duality Total Recall - Tutorial," YouTube, Dec. 12, 2013. Accessed: Aug. 13, 2026. [Online video]. Available: <https://www.youtube.com/watch?v=Bu9bETpqT60>
- [12] E. Odunlade, "Top 10 popular microcontrollers among makers," *Electronics-Lab*, June 19, 2020. Accessed: Aug. 13, 2026. [Online]. Available: <https://www.electronics-lab.com/top-10-popular-microcontrollers-among-makers/>
- [13] "Rotary encoder on a single MCU pin," Elektor Magazine. Accessed: Aug. 13, 2026. [Online]. Available: <https://www.elektormagazine.com/labs/rotary-encoder-on-a-single-mcu-pin>

What are your questions?

Github Repository



https://github.com/skylar-slideshows/Digilog_Capstone